

get()

```
char str[ 10 ];  
for( int i = 0; i < 3; i++ )  
    inFile.get( str, 10, '\n');
```

str

--	--	--	--	--	--	--	--	--	--

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

```
char str[ 10 ];
for( int i = 0; i < 3; i++ )
    inFile.get( str, 10, '\n');
```

str

--	--	--	--	--	--	--	--	--	--

streambuf

a	b	c		d	e	\n	f	g	h	i		j	k	l	\n		
---	---	---	--	---	---	----	---	---	---	---	--	---	---	---	----	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
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```
char str[ 10 ];
for( int i = 0; i < 3; i++ )
    inFile.get( str, 10, '\n');
```

str

a	b	c		d	e	\0			
---	---	---	--	---	---	----	--	--	--

streambuf

\n	f	g	h	i		j	k	l	\n								
----	---	---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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```
char str[ 10 ];  
for( int i = 0; i < 3; i++ )  
    inFile.get( str, 10, '\n');
```

str

a	b	c		d	e	\0			
---	---	---	--	---	---	----	--	--	--

streambuf

\n	f	g	h	i		j	k	l	\n								
----	---	---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	false
<code>inFile.eof()</code>	false
<code>inFile.fail()</code>	true
<code>inFile.bad()</code>	false

getline()

```
char str[ 10 ];  
for( int i = 0; i < 3; i++ )  
    inFile.getline( str, 10, '\n');
```

str

--	--	--	--	--	--	--	--	--	--

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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<code>inFile.bad()</code>	<code>false</code>

```
char str[ 10 ];
for( int i = 0; i < 3; i++ )
    inFile.getline( str, 10, '\n');
```

str

--	--	--	--	--	--	--	--	--	--

streambuf

a	b	c		d	e	\n	f	g	h	i		j	k	l	\n		
---	---	---	--	---	---	----	---	---	---	---	--	---	---	---	----	--	--



abc de\n
fghi jkl\n

File

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false


```
char str[ 10 ];
for( int i = 0; i < 3; i++ )
    inFile.getline( str, 10, '\n');
```

str

a	b	c		d	e	\0			
---	---	---	--	---	---	----	--	--	--

streambuf

\n	f	g	h	i		j	k	l	\n								
----	---	---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
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```
char str[ 10 ];
for( int i = 0; i < 3; i++ )
    inFile.getline( str, 10, '\n');
```

str

a	b	c		d	e	\0			
---	---	---	--	---	---	----	--	--	--

streambuf

f	g	h	i		j	k	l	\n									
---	---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
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```
char str[ 10 ];
for( int i = 0; i < 3; i++ )
    inFile.getline( str, 10, '\n');
```

str

f	g	h	i		j	k	l	\0	
---	---	---	---	--	---	---	---	----	--

streambuf

\n																	
----	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

```
char str[ 10 ];
for( int i = 0; i < 3; i++ )
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```

str

f	g	h	i		j	k	l	\0	
---	---	---	---	--	---	---	---	----	--

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

```
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for( int i = 0; i < 3; i++ )
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```

str

f	g	h	i		j	k	l	\0	
---	---	---	---	--	---	---	---	----	--

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



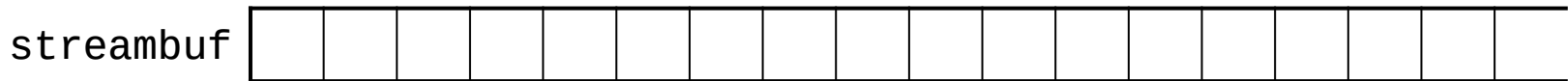
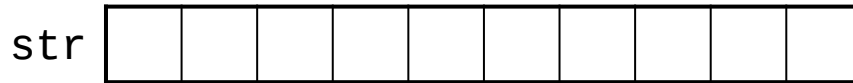
abc de\n
fghi jkl\n

File

inFile.good()	false
inFile.eof()	true
inFile.fail()	true
inFile.bad()	false

Stream Extraction Operator >>

```
char str[ 10 ];  
for( int i = 0; i < 5; i++ )  
    inFile >> str;
```



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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<code>inFile.bad()</code>	<code>false</code>

```
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```

str

--	--	--	--	--	--	--	--	--	--

streambuf

a	b	c		d	e	\n	f	g	h	i		j	k	l	\n		
---	---	---	--	---	---	----	---	---	---	---	--	---	---	---	----	--	--



abc de\n
fghi jkl\n

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<code>inFile.good()</code>	<code>true</code>
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```
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for( int i = 0; i < 5; i++ )
    inFile >> str;
```

str

a	b	c	\0						
---	---	---	----	--	--	--	--	--	--

streambuf

d	e	\n	f	g	h	i		j	k	l	\n					
---	---	----	---	---	---	---	--	---	---	---	----	--	--	--	--	--



abc de\n
fghi jkl\n

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```

str

d	e	\0							
---	---	----	--	--	--	--	--	--	--

streambuf

\n	f	g	h	i		j	k	l	\n								
----	---	---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

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for( int i = 0; i < 5; i++ )  
    inFile >> str;
```

str

d	e	\0							
---	---	----	--	--	--	--	--	--	--

streambuf

f	g	h	i		j	k	l	\n									
---	---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

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```
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for( int i = 0; i < 5; i++ )
    inFile >> str;
```

str

f	g	h	i	\0					
---	---	---	---	----	--	--	--	--	--

streambuf

j	k	l	\n														
---	---	---	----	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

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str

j	k	l	\0						
---	---	---	----	--	--	--	--	--	--

streambuf

\n																	
----	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

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for( int i = 0; i < 5; i++ )  
    inFile >> str;
```

str

j	k	l	\0						
---	---	---	----	--	--	--	--	--	--

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

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<code>inFile.good()</code>	<code>true</code>
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str

j	k	l	\0						
---	---	---	----	--	--	--	--	--	--

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>false</code>
<code>inFile.eof()</code>	<code>true</code>
<code>inFile.fail()</code>	<code>true</code>
<code>inFile.bad()</code>	<code>false</code>

get()


```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c \0

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c

\0

streambuf

a	b	c		d	e	\n	f	g	h	i		j	k	l	\n		
---	---	---	--	---	---	----	---	---	---	---	--	---	---	---	----	--	--



abc de\n
fghi jkl\n

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<code>inFile.good()</code>	<code>true</code>
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<code>inFile.bad()</code>	<code>false</code>

```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c

a

streambuf

b	c		d	e	\n	f	g	h	i		j	k	l	\n			
---	---	--	---	---	----	---	---	---	---	--	---	---	---	----	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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<code>inFile.bad()</code>	<code>false</code>

```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c

b

streambuf

c		d	e	\n	f	g	h	i		j	k	l	\n				
---	--	---	---	----	---	---	---	---	--	---	---	---	----	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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<code>inFile.bad()</code>	<code>false</code>

```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c

c

streambuf

	d	e	\n	f	g	h	i		j	k	l	\n				
--	---	---	----	---	---	---	---	--	---	---	---	----	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
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<code>inFile.bad()</code>	<code>false</code>

```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c

k

streambuf

l	\n																
---	----	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c l

streambuf

\n																	
----	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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<code>inFile.bad()</code>	<code>false</code>

```
char c = '\0';  
for( int i = 0; i < 17; i++ )  
    c = inFile.get();
```

c \n

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
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    c = inFile.get();
```

c \n

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abc de\n
fghi jkl\n

File

<code>inFile.good()</code>	false
<code>inFile.eof()</code>	true
<code>inFile.fail()</code>	true
<code>inFile.bad()</code>	false

Stream Extraction Operator >>
followed by `get()`

```
char str[ 10 ];  
inFile >> str;  
inFile.get( str, 10, '\n' );
```

str

--	--	--	--	--	--	--	--	--	--

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



abcde\n
fghijkl\n

File

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

```
char str[ 10 ];  
inFile >> str;  
inFile.get( str, 10, '\n' );
```

str

--	--	--	--	--	--	--	--	--	--

streambuf

a	b	c	d	e	\n	f	g	h	i	j	k	l	\n				
---	---	---	---	---	----	---	---	---	---	---	---	---	----	--	--	--	--



abcde\n
fghijkl\n

File

inFile.good()	true
inFile.eof()	false
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```
char str[ 10 ];  
inFile >> str;  
inFile.get( str, 10, '\n' );
```

str

a	b	c	d	e	\0				
---	---	---	---	---	----	--	--	--	--

streambuf

\n	f	g	h	i	j	k	l	\n									
----	---	---	---	---	---	---	---	----	--	--	--	--	--	--	--	--	--



abcde\n
fghijkl\n

File

inFile.good()	true
inFile.eof()	false
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```
char str[ 10 ];  
inFile >> str;  
inFile.get( str, 10, '\n' );
```

str

a	b	c	d	e	\0				
---	---	---	---	---	----	--	--	--	--

streambuf

\n	f	g	h	i	j	k	l	\n									
----	---	---	---	---	---	---	---	----	--	--	--	--	--	--	--	--	--



abcde\n
fghijkl\n

File

<code>inFile.good()</code>	<code>false</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>true</code>
<code>inFile.bad()</code>	<code>false</code>

Stream Extraction Operator >>

```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



100	101\n
102	103\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>


```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number

streambuf

1	0	0		1	0	1	\n	1	0	2		1	0	3	\n		
---	---	---	--	---	---	---	----	---	---	---	--	---	---	---	----	--	--



100	101\n
102	103\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number 100

streambuf

1	0	1	\n	1	0	2		1	0	3	\n						
---	---	---	----	---	---	---	--	---	---	---	----	--	--	--	--	--	--



100	101\n
102	103\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number 101

streambuf

\n	1	0	2		1	0	3	\n									
----	---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--	--



100 101\n

102 103\n

File

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number 101

streambuf

1	0	2		1	0	3	\n										
---	---	---	--	---	---	---	----	--	--	--	--	--	--	--	--	--	--



100 101\n
102 103\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number 102

streambuf

1	0	3	\n													
---	---	---	----	--	--	--	--	--	--	--	--	--	--	--	--	--



100 101\n

102 103\n

File

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number 103

streambuf

\n																	
----	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



100 101\n
102 103\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

```
int number;  
for( int i = 0; i < 5; i++ )  
    inFile >> number;
```

number 103

streambuf

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



100 101\n
102 103\n

File

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

If the state of streambuf is good, and the first character in streambuf is not '`\n`', then after the execution of

- `inFile.get`, '`\n`' is not removed from streambuf,
- `inFile.getline`, '`\n`' is removed from streambuf if found,
- `inFile.getline`, the state of streambuf is set to fail if '`\n`' is not found,
- `inFile >>`, '`\n`' is not removed from streambuf.

If the state of streambuf is good, and the first character in streambuf is '`\n`', then the execution of

- `inFile.get` leaves '`\n`' in streambuf, reads nothing and changes the state to fail,
- `inFile.getline` removes '`\n`' from streambuf, reads an empty string and keeps the state good,
- `inFile >>` removes '`\n`' from streambuf, do correct read operation and keeps the state good.